Word Problems & Reasoning

ClipSAT Printable Practice Suite (clipsat.org) • Topic 8.01

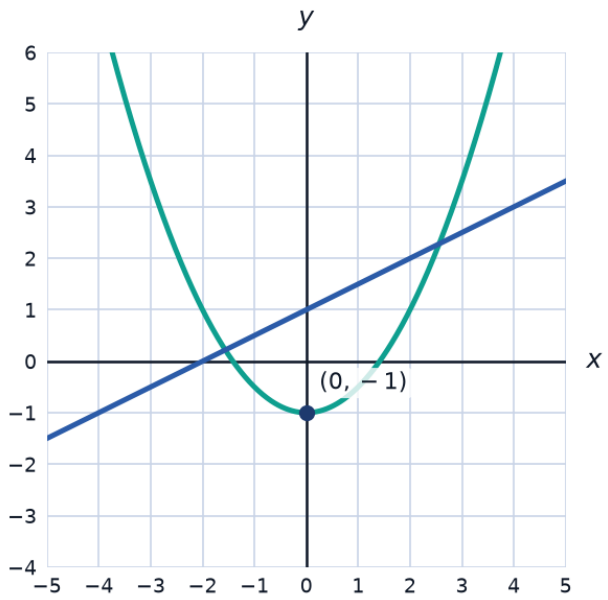
Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational] • Target: Number and digit reasoning**

A number decreased by 9 equals 34. Find the number.

2 **[Foundational] • Target: Number and digit reasoning**

Three times a number, minus 7, equals 32. Find the number.



3 **[Foundational] • Target: Number and digit reasoning**

The sum of two numbers is 58. One number is 12 more than the other. Find both numbers.

4 **[Foundational] • Target: Number and digit reasoning**

A two-digit number has digit sum 11. Reversing its digits gives a number 45 more than the original. Find the original number.

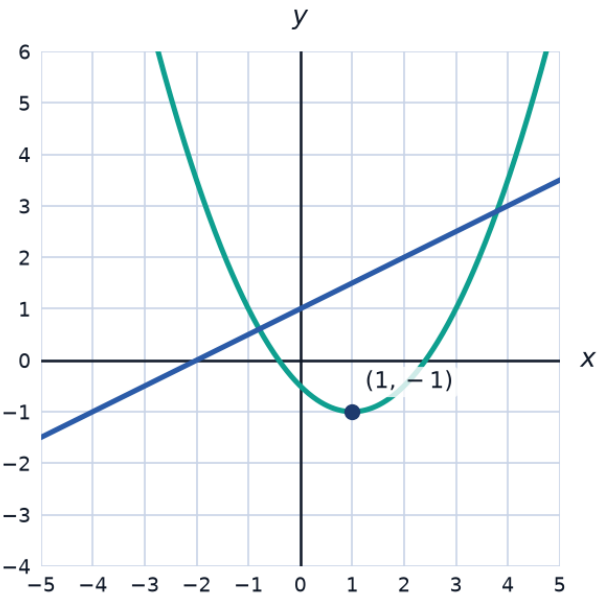
5 **[Foundational] • Target: Age problems**

A father is 4 times as old as his daughter. In 5 years, he will be 3 times as old as she will be. Find their current ages.

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6 [Foundational] • Target: Age problems

Sara is twice as old as her brother. In 6 years, the sum of their ages will be 42. Find their current ages.



7 [Foundational] • Target: Age problems

Ahmed is 5 years older than Salem. Seven years ago, Ahmed was twice as old as Salem. Find their current ages.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Age problems

The sum of a mother's and daughter's ages is 50. The mother is 20 years older than the daughter. Find both ages.

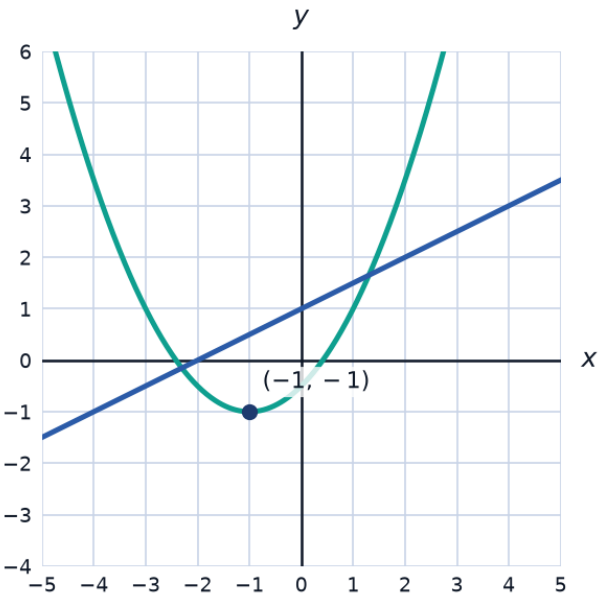
9 [Intermediate] • Target: Distance, rate and time

A train travels 180 km in 2 hours. At the same speed, how far does it travel in 5 hours?

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10 [Intermediate] • Target: Distance, rate and time

Two cities are 350 km apart. A car leaves one city for the other at 70 km/h. How long does the trip take?



11 [Intermediate] • Target: Distance, rate and time

Two cyclists start at the same point and ride in opposite directions, at 15 km/h and 20 km/h. After how many hours will they be 105 km apart?

12 [Intermediate] • Target: Distance, rate and time

A plane flies against a headwind at 480 km/h and covers 1,440 km. How many hours does the flight take?

13 [Intermediate] • Target: Work-rate problems

Worker A can paint a room in 6 hours; Worker B can paint the same room in 3 hours. Working together, how long will it take them?

14 [Intermediate] • Target: Work-rate problems

A pipe fills a tank in 10 hours; a second pipe fills it in 15 hours. Working together, how long to fill the tank?

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

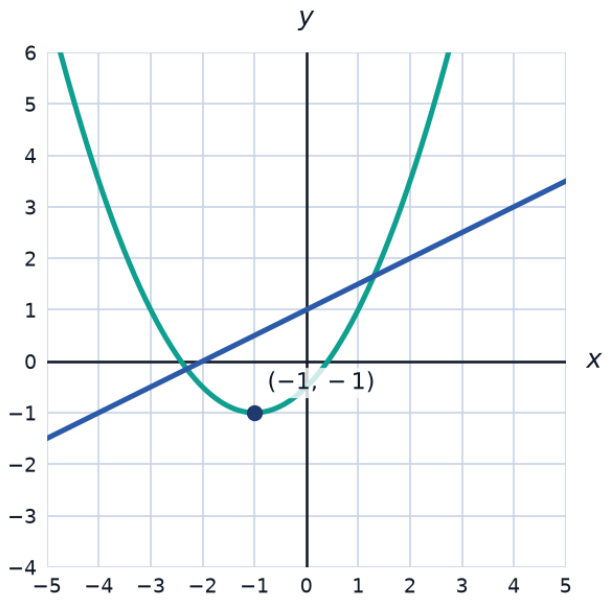
15 [Challenging] • Target: Work-rate problems • Cross-Lesson: 2.01 Ratios, Proportions & Percentages

6 workers can complete a job in 8 days. How many days would 4 workers take, at the same rate?

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16 [Challenging] • Target: Work-rate problems • Cross-Lesson: 3.01 Algebra Essentials

A pipe fills a tank in 12 hours, but a leak can empty a full tank in 24 hours. If both act at once, how long to fill the tank?



17 [Challenging] • Target: Mixture and money problems • Cross-Lesson: 4.01 Geometry & Measurement

A jeweler mixes gold worth 200 riyals/gram with silver worth 50 riyals/gram to make 10 grams of a mixture worth 110 riyals/gram. How many grams of gold are used?

18 [Challenging] • Target: Mixture and money problems • Cross-Lesson: 5.01 Quantitative Comparisons

A shopkeeper has 15 riyals in 1-riyal and 2-riyal coins, 10 coins in total. How many of each coin does she have?

19 [Challenging] • Target: Mixture and money problems • Cross-Lesson: 6.01 Sequences & Patterns

A solution is 40% acid. How many liters of pure water must be added to 20 liters of this solution to dilute it to 25% acid?

20 [Challenging] • Target: Mixture and money problems • Cross-Lesson: 7.01 Data Analysis & Probability

Event tickets cost 30 riyals for adults and 18 riyals for children. 150 tickets were sold for 3,780 riyals in total. How many adult tickets were sold?

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

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21 [Foundational] • Target: MCQ — Number and digit reasoning

A number decreased by 12 equals 47. Find the number.

a. 59

b. 65

c. 35

d. 47

22 [Foundational] • Target: MCQ — Number and digit reasoning

A two-digit number has digit sum 9. Reversing its digits gives a number 27 more than the original. Find the original number.

a. 18

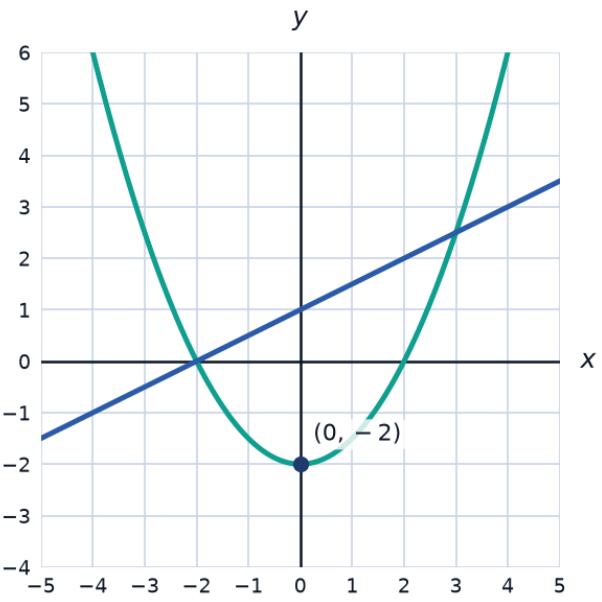
b. 36

c. 63

d. 45

23 [Foundational] • Target: MCQ — Age problems

Huda is twice as old as her sister. In 8 years, the sum of their ages will be 46. Find Huda's current age.



a. 10

b. 18

c. 14

d. 20

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24 [Foundational] • Target: MCQ — Age problems

Tariq is currently twice as old as his son. If Tariq is 40, find his son's age.

a. 18

b. 20

c. 22

d. 80

25 [Foundational] • Target: MCQ — Distance, rate and time

A plane flies against a headwind at 400 km/h and covers 1,600 km. How many hours does the flight take?

a. 4,000

b. 4

c. 5

d. 400

26 [Foundational] • Target: MCQ — Work-rate problems

8 workers can complete a job in 6 days. How many days would 12 workers take, at the same rate?

a. 4

b. 9

c. 3

d. 6

27 [Foundational] • Target: MCQ — Mixture and money problems

A shopkeeper has 21 riyals in 1-riyal and 2-riyal coins, 15 coins in total. How many 2-riyal coins does he have?

a. 9

b. 15

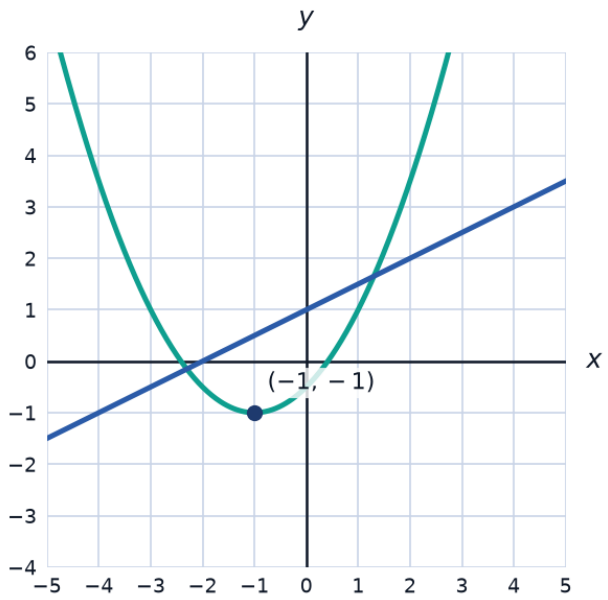
c. 6

d. 3

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28 [Foundational] • Target: MCQ — Logical deduction

Omar, Bandar and Nasser have 44 books together. Omar has twice as many as Bandar, and Nasser has 8 more than Bandar. Find how many books Bandar has.



a. 22

b. 17

c. 9

d. 18

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: MCQ — Logical deduction

Three consecutive integers sum to 72. Find the smallest one.

a. 25

b. 24

c. 23

d. 22

30 [Intermediate] • Target: MCQ — Working backward

After a 20% pay cut, an employee's salary is 3,200 riyals. Find the salary before the cut.

a. 4,200

b. 4,000

c. 3,600

d. 3,840

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31 [Intermediate] • Target: MCQ — Mixed word problems

Sarah buys 4 notebooks and 3 pens for 38 riyals. Khalid buys 2 notebooks and 5 pens for 40 riyals, at the same prices. Find the price of one notebook.

a. 6

b. 8

c. 5

d. 4

32 [Intermediate] • Target: MCQ — Mixed word problems

A cyclist travels 60 km at a certain speed, then returns the same 60 km at half that speed, taking 9 hours total. Find the original speed.

a. 15

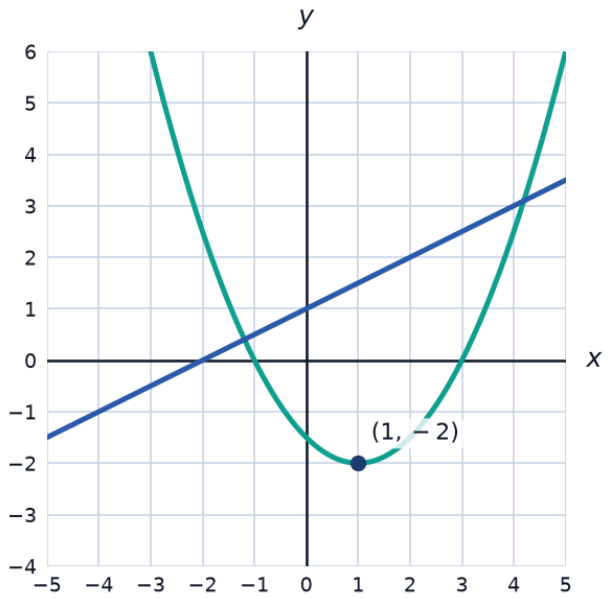
b. 10

c. 40

d. 20

33 [Intermediate] • Target: MCQ — More rate and proportion problems

A factory produces 1,500 items in 6 hours. At the same rate, how many items would it produce in a 9-hour shift?



a. 1,875

b. 2,700

c. 2,250

d. 2,000

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34 [Intermediate] • Target: MCQ — Work-rate problems

A number decreased by 15 equals 26. Find the number.

a. 41 **b.** 39

c. 46 **d.** 43

35 [Intermediate] • Target: MCQ — Mixture and money problems

A number decreased by 13 equals 66. Find the number.

a. 79 **b.** 77

c. 84 **d.** 81

36 [Intermediate] • Target: MCQ — Logical deduction

A number decreased by 38 equals -19. Find the number.

a. 21 **b.** 19

c. 17 **d.** 24

37 [Intermediate] • Target: MCQ — Working backward

Three times a number, minus 13, equals 17. Find the number.

a. 15 **b.** 8

c. 12 **d.** 10

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: MCQ — Mixed word problems • Cross-Lesson: 5.01 Quantitative Comparisons

A train travels 535 km in 5 hours. At the same speed, how far does it travel in 2 hours?

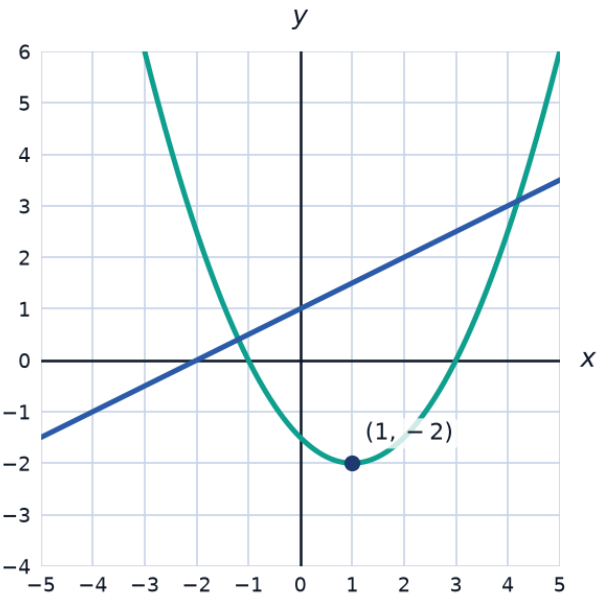
a. 219 **b.** 212

c. 214 **d.** 216

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39 [Challenging] • Target: MCQ — More rate and proportion problems • Cross-Lesson: 6.01 Sequences & Patterns

Two cities are 294 km apart. A car leaves one city for the other at 49 km/h. How long does the trip take?



a. 4

b. 8

c. 11

d. 6

40 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 7.01 Data Analysis & Probability

A train travels 202 km in 2 hours. At the same speed, how far does it travel in 5 hours?

a. 510

b. 503

c. 507

d. 505

41 [Challenging] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14) • Cross-Lesson: 1.01 Arithmetic & Number Sense

A train travels 540 km in 6 hours. At the same speed, how far does it travel in 8 hours?

a. 720

b. 725

c. 718

d. 722

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42 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 2.01 Ratios, Proportions & Percentages

Yousef is 5 times as old as his younger sibling. In 6 years, the sum of their ages will be 60. Find their current ages.

a. 8, 40

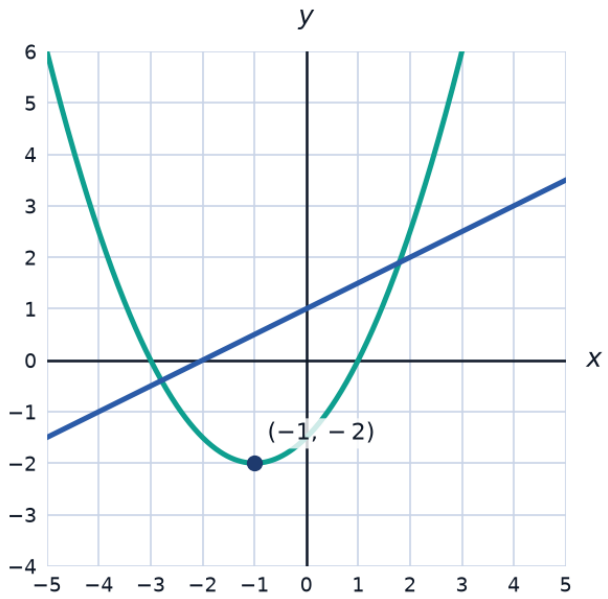
b. 8, 44

c. 10, 40

d. 40, 8

43 [Challenging] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28) • Cross-Lesson: 3.01 Algebra Essentials

Worker A can complete a job in 12 hours; Worker B can complete the same job in 4 hours. Working together, how long will it take them?



a. 8

b. 3

c. 2

d. 4

44 [Challenging] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37) • Cross-Lesson: 4.01 Geometry & Measurement

Worker A can complete a job in 8 hours; Worker B can complete the same job in 24 hours. Working together, how long will it take them?

a. 7

b. 6

c. 16

d. 5

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45 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 5.01 Quantitative Comparisons

After spending half her money on a book, then 13 riyals on lunch, Huda has 18 riyals left. How much did she start with?

a. 72

b. 62

c. 18

d. 52

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